

Mohamed Aamir

PhD Scholar — Ocean Engineering
Indian Institute of Technology Madras

 LinkedIn —  GitHub —  ResearchGate —  ORCID

Profile

Sophomore PhD scholar in the Department of Ocean Engineering at IIT Madras, working under the guidance of Dr. K. Narendran. Research interests include hydrodynamics, machine learning, numerical modeling, CFD, and computational analysis of offshore systems.

Education

Indian Institute of Technology Madras MS & PhD in Ocean Engineering CGPA: 9.52	Present
Mohamed Sathak Engineering College, Kilakarai B.E. in Civil Engineering CGPA: 8.56	2020 – 2024

Research Focus

Machine Learning for Computational Fluid Dynamics

My research focuses on the application of machine learning techniques to computational engineering problems.

Teaching Experience

Module Instructor: Introduction to OpenFOAM Conducted the OpenFOAM module for <i>OE5555: Summer Training in Computer Modeling and Simulation</i> . Instructed Master's students on CFD workflows, mesh generation, solver setup, and hydrodynamic simulations.	May 2025
--	----------

Academic Service

Journal Reviewing

- Physics of Fluids (AIP Publishing), Reviewer, 2026

Internship Experience

Structural Engineering Intern Almino Structural Consultancy, Ramanathapuram • Assisted in structural drafting and detailing using AutoCAD • Performed rebar checking and site verification activities	Jan – May 2024
---	----------------

Selected Coursework

- Numerical Techniques in Ocean Hydrodynamics
- Experimental Methods and Measurements
- Installation of Offshore Structures
- Wave Hydrodynamics
- Machine Learning for Ocean Engineers
- Pipeline & Riser Engineering
- Coastal Engineering
- Advanced Dynamics of Floating Bodies

Technical Skills

- OpenFOAM
- CFD and Numerical Modeling
- Hydrodynamic Analysis
- Python
- Machine Learning
- MATLAB
- AutoCAD

Research Outputs

Manuscripts and research outputs currently under preparation.

Mohamed Aamir
Department of Ocean Engineering, IIT Madras